

Free-Space Optical Communication using Photonic-Crystal Surface-Emitting Lasers

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Abstract Photonic-crystal surface-emitting lasers (PCSELs) offer a compelling route to compact free-space optical (FSO) transmitters by combining watt-class single-mode output, narrow divergence, and direct modulation on a single chip. This invited paper reviews recent progress from initial lens-free demonstrations to >10-Gbit/s, and coherent PCSEL-based links. ©2026 The Author(s)

Introduction

Free-space optical (FSO) communication has been attracting increasing attention for future terrestrial and space networks because it enables high-capacity links without fiber deployment and offers flexibility for fronthaul, backhaul, inter-satellite, and deep-space applications [1,2]. At the same time, practical FSO systems require transmitters that combine high output power, narrow beam divergence, and high-speed modulation in a compact and efficient form [3]. Conventional semiconductor lasers do not easily satisfy all these requirements on a single chip, and therefore practical transmitters often rely on bulky combinations of laser sources, fiber amplifiers, external modulators, and beam-shaping optics.

Photonic-crystal surface-emitting lasers (PCSELs) provide a promising alternative. Since they exploit two-dimensional coherent resonance in a photonic-crystal cavity, PCSELs can realize single-mode lasing over a much larger area than conventional semiconductor lasers [4,5]. This unique property enables both watt-class output power and narrow beam divergence, while preserving the compactness of semiconductor

devices. Double-lattice photonic-crystal designs with non-Hermitian coupling control have further strengthened this platform by enabling large-area single-mode emission with high brightness [6-8].

Recent progress has shown a clear evolution of PCSEL-based FSO transmitters (Fig. 1). Initial work demonstrated direct-modulation FSO links using PCSELs and revealed the feasibility of lens-free transmission [9]. Later work established watt-class continuous-wave (CW) PCSELs with several-GHz direct modulation, enabling over 10-Gbit/s transmission and virtual kilometer-class links without a transmitter lens and fiber amplifier [10]. Most recently, two-section PCSELs capable of watt-class frequency modulation have opened the way to compact coherent FSO links with link budgets exceeding 80 dB [11,12]. Also, we have advanced PCSEL-based communication through various concepts, including bandwidth expansion via photon-photon resonance (PPR) [13,14] and high-extinction-ratio signal generation.

This invited paper reviews that progression and discusses why PCSELs are emerging as a key transmitter technology for compact FSO systems.

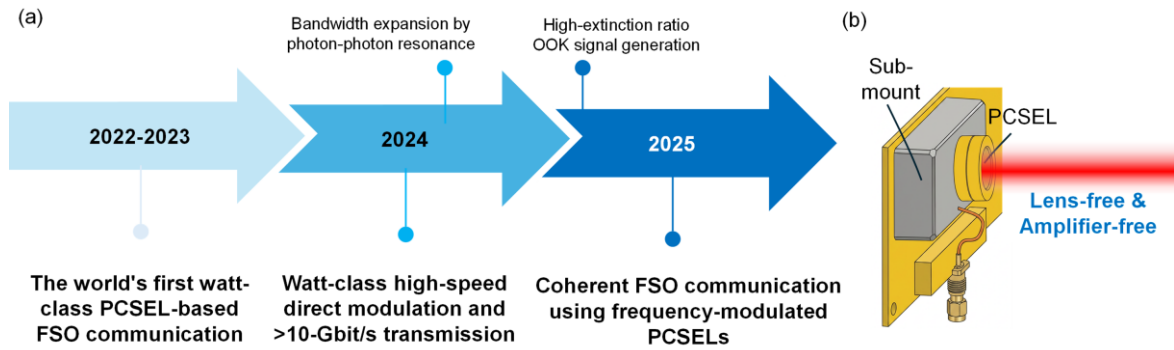


Fig. 1: (a) Recent advances in PCSEL-based FSO communication. (b) PCSEL-based transmitter configuration.

Why PCSELS are promising for FSO transmitters

The main advantage of a PCSEL is that it combines a large coherent emission area with single-mode oscillation. In conventional edge-emitting lasers or vertical-cavity surface-emitting lasers (VCSELs), increasing the emission aperture generally leads to deteriorated beam quality and/or multimode behavior. As a result, conventional FSO transmitters usually require additional optics to collimate the beam and optical amplifiers to obtain sufficiently high launched power. By contrast, PCSELS can emit high-power, narrow divergence beams directly from the chip itself as shown in Fig. 1(b). This feature is particularly attractive for FSO communication. First, the narrow divergence reduces beam spreading and simplifies transmitter optics. Second, the watt-class output power potentially removes the need for fiber amplifiers. Third, if the device can be modulated directly at GHz-class bandwidths, a single PCSEL chip can play the role of source, power booster, beam former, and modulator simultaneously. These properties are especially beneficial in space communication applications, where size, weight, and power consumption are critical.

Another important advantage is coherence. Recent studies have shown that large-area PCSELS can exhibit narrow intrinsic linewidths and can be adapted for coherent modulation [15]. Therefore, the same device platform is relevant not only to intensity-modulated/direct-detection (IM/DD) links but also to coherent FSO systems aimed at longer reach and higher sensitivity.

First demonstrations of PCSEL-based FSO links

The first FSO communication experiments using 940-nm PCSELS demonstrated the basic feasibility of the concept [9]. In that study, a 500- μm -diameter PCSEL was employed under quasi-CW (QCW) operation and provided more than 1 W peak power. Using this source, lens-free transmission over 1.1 m was demonstrated with 480-MHz and 864-MHz 64QAM OFDM signals, corresponding to gross rates of 2.88 Gb/s and 5.18 Gb/s, respectively. Although the setup was still limited by QCW operation and RF delivery issues, the work clearly showed that PCSELS can simplify the transmitter by eliminating the need for a transmitter lens.

Watt-class high-speed direct modulation and >10-Gbit/s transmission

The next step was to determine whether high output power and high-speed direct modulation could be achieved simultaneously under CW operation. For this purpose, the PCSEL modulation response was analyzed in terms of intrinsic

optical dynamics and parasitic electrical effects [10]. The study showed that watt-class PCSELS can retain several-GHz-class modulation bandwidth when their active region, photonic-crystal cavity, and package parasitic are properly designed.

Based on these considerations, a 500- μm -diameter PCSEL mounted on a cooling package was fabricated. The device achieved watt-class CW operation at ~ 3 A injection current while maintaining single-mode lasing. Frequency-response measurements demonstrated several-GHz direct modulation at watt-class operating points, confirming that high power and high speed are compatible in this platform.

Using this PCSEL, FSO transmission experiments were performed without a transmitter lens. Quadrature-amplitude-modulated (QAM) signals were directly imposed on the laser, and link loss was emulated by optical attenuation before detection. At 2 Gbaud, 64QAM transmission achieved a gross rate of 12 Gbit/s, while 256QAM transmission reached 16 Gbit/s gross. Under a distance-conversion model assuming a 20-cm-diameter receiver lens, the 16-Gbit/s case at 20 dB attenuation corresponded to approximately 500 m. In addition, QPSK transmission at 1 Gbaud under 38 dB attenuation corresponded to a virtual transmission distance of approximately 5 km. With 20% soft-decision FEC overhead, the 16-Gbit/s gross rate corresponds to 13.3 Gbit/s net throughput. We note that this link budget could be further improved by using an avalanche photodiode (APD).

These results considerably strengthened the case for PCSEL-based transmitters. They demonstrated that a single directly modulated watt-class PCSEL can simultaneously provide high optical power, narrow divergence, and multi-Gbit/s communication performance without a transmitter lens and fiber amplifier. This represents a major architectural simplification compared with conventional FSO transmitters.

Coherent FSO communication using frequency-modulated PCSELS

A further major development was the extension of the PCSEL platform to coherent communication [11,12]. In the most recent work, a two-section PCSEL was proposed and demonstrated for high-power frequency modulation with suppressed amplitude modulation [12]. The device integrates two photonic-crystal regions with slightly different band-edge resonant frequencies. By applying antiphase RF currents to the two sections, the lasing frequency can be modulated efficiently while suppressing unwanted intensity modulation.

Experimentally, the fabricated two-section 500- μm -diameter PCSEL delivered more than 1

W under CW operation at 3 A bias and exhibited a narrow beam divergence of about 0.2° . Under antiphase RF modulation near 1 GHz, a frequency excursion of approximately 0.55 GHz peak-to-peak was obtained while keeping the amplitude modulation depth quite low. Compared with a conventional single-section device, the two-section PCSEL showed a much better ratio of frequency modulation to amplitude modulation, confirming its suitability as a coherent optical transmitter.

The same study then demonstrated proof-of-concept coherent FSO communication using heterodyne detection with another reference PCSEL. Binary non-return-to-zero (NRZ) signals were transmitted, and the optical signal was attenuated to emulate long-distance propagation. At 0.5 Gbaud, a clear eye diagram was observed even at a received optical power of -50 dBm. The measured minimum allowable received powers were -58 dBm at 0.5 Gbaud and -51 dBm at 1.0 Gbaud under a 20% soft-decision FEC criterion, corresponding to allowable power attenuations of 88 dB and 81 dB, respectively.

These link budgets are highly significant because they approach the range relevant to long-distance space optical links. Equally important, the transmitter operated without an optical amplifier. In other words, the PCSEL itself generated the coherent watt-class optical signal, eliminating the need for the bulky combination of semiconductor laser, fiber amplifier, and external phase modulator typically used in coherent FSO systems. This result suggests a path toward one-chip coherent transmitters with drastically reduced size and weight.

Conclusion

In conclusion, PCSELS are emerging as a key transmitter technology for compact FSO communication systems. Their ability to provide watt-class output power, narrow beam divergence, single-mode operation, and GHz-class modulation on a single chip enables a substantial simplification of conventional FSO transmitter architectures. As reviewed in this paper, PCSEL-based FSO links have progressed from initial lens-free transmission demonstrations to multi-Gbit/s direct-modulation links under CW operation, and more recently to coherent communication using two-section frequency-modulated PCSELS with link budgets exceeding 80 dB. In addition, recent progress has also been made in FSO transmission at the eye-safe C-band using InP-based PCSELS, further extending the application range of this technology [16]. These results highlight the strong potential of PCSELS for both IM/DD and coherent FSO systems, particularly in applications requiring compactness, low weight, and high performance.

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